```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay


# Load the dataset
df = pd.read_csv("diabetes.csv")


# Remove missing values
df.dropna(inplace=True)


# Separate features and target
x = df.drop("Outcome", axis=1)
y = df["Outcome"]


# Split the dataset into training and testing sets
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
# Logistic Regression without feature scaling
model_2 = LogisticRegression(max_iter=10000)
model_2.fit(x_train, y_train)
pred_2 = model_2.predict(x_test)
accuracy_2 = accuracy_score(y_test, pred_2)
precision_2 = precision_score(y_test, pred_2)
recall_2 = recall_score(y_test, pred_2)
f1_score_2 = f1_score(y_test, pred_2)


print("\n---- Logistic Regression Without Feature Scaling ----")
print("\n---- MODEL PERFORMANCE ----")
print("Accuracy  : ", accuracy_2)
print("Precision : ", precision_2)
print("Recall    : ", recall_2)
print("F1 Score  : ", f1_score_2)
# Confusion matrix
cm = confusion_matrix(y_test, pred_2)
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["No Diabetes", "Diabetes"]
)
disp.plot()
plt.title("Without Feature Scaling")
plt.show()
# Apply feature scaling
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
# Logistic Regression with feature scaling
model_1 = LogisticRegression(max_iter=10000)
model_1.fit(x_train, y_train)


pred_1 = model_1.predict(x_test)
accuracy_1 = accuracy_score(y_test, pred_1)
precision_1 = precision_score(y_test, pred_1)
recall_1 = recall_score(y_test, pred_1)
f1_score_1 = f1_score(y_test, pred_1)


print("\n---- Logistic Regression With Feature Scaling ----")
print("\n---- MODEL PERFORMANCE ----")
print("Accuracy  : ", accuracy_1)
print("Precision : ", precision_1)
print("Recall    : ", recall_1)
print("F1 Score  : ", f1_score_1)
# Confusion matrix
cm = confusion_matrix(y_test, pred_1)
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["No Diabetes", "Diabetes"] )
disp.plot()
plt.title("With Feature Scaling")
plt.show()
```

```
---- Logistic Regression Without Feature Scaling -

---- MODEL PERFORMANCE ----
Accuracy  :  0.7467532467532467
Precision :  0.6379310344827587
Recall    :  0.6727272727272727
F1 Score  :  0.6548672566371682

---- Logistic Regression With Feature Scaling ----

---- MODEL PERFORMANCE ----
Accuracy  :  0.7532467532467533
Precision :  0.6491228070175439
Recall    :  0.6727272727272727
F1 Score  :  0.6607142857142857
```



